

Unit Test -2083

Subject: Computer
Grade: 10

Time: 2 Hours

Full Marks: 50
Pass Marks: 18

Q.1 Choose the correct answer: (6×1=6)

- a) **Telecommunications mainly deals with:**
 - i) Storing data in computers
 - ii) Repairing hardware
 - iii) Sending and receiving data over long distances
 - iv) Designing software
- b) **Data packets are:**
 - i) Large storage devices
 - ii) Antivirus Program
 - iii) Small pieces of data sent over a network
 - iv) Hardware chips
- c) **Bandwidth is:**
 - i) Actual data speed
 - ii) Delay in transmission
 - iii) Maximum data capacity of a network
 - iv) Variation in signal
- d) **5g technology offers:**
 - i) Lower speed than 4g
 - ii) Only voice calls
 - iii) Very high speed and low latency
 - iv) Wired connectivity only
- e) **Rj45 connector is commonly used with:**
 - i) Fiber optic cable
 - ii) Coaxial cable
 - iii) Twisted pair cable
 - iv) Satellite dish
- f) **A media converter works at:**
 - i) Physical Layer
 - ii) Network Layer
 - iii) Application layer
 - iv) Transport layer

Q.2 Give Technical terms (6×1=6)

- a) Cable with a central conductor, insulating layer, and shield.
- b) A device that controls two dissimilar networks.
- c) Network of networks.
- d) Privately owned network of computers.
- e) Sending and receiving messages electronically via the Internet.
- f) An interconnected pattern of network components.

Q.3 Write full forms of the following (6×1=6)

- a) SMTP
- b) IPX/SPX
- c) POP
- d) FTP
- e) HTTP
- f) URL

Q.4 Write short answers. (10×1=10)

- a) Define computer network.
- b) Write any two advantages of a computer network.
- c) Write one use of fiber optic cable.
- d) Write any two examples of protocols.
- e) Define Extranet.
- f) What is meant by database?
- g) What is an IP address ?
- h) What is the role of switch?
- i) Define the Internet in a sentence.
- j) Write two examples of web browsers.

Q.5 Write long answers. (6×2=12)

- a) Describe the role of satellites in Communication.
- b) What is a computer network? Write any two advantages and limitations of it.
- c) Differentiate between guided and unguided communication channels.
- d) Explain the use of Cat 6 cable.
- e) Explain the difference between simplex, half-duplex, and full-duplex data transmission modes.
- f) What is a MAC address?

Q.6 Write SQL commands for the following questions. (10)

Student

StudentID	Name	Age	Address
101	Aayush Sharma	19	Pokhara
102	Sita Karki	20	Kathmandu
103	Rohan Gurung	20	Butwal
104	Priya Rai	21	Dharan
105	Nabin Thapa	19	Chitwan

Employee

EmpID	EmpName	Salary	Department
201	Ram Bhandari	35000	HR
202	Anita Sharma	42000	IT
203	Suman KC	38000	Finance
204	Deepa Magar	40000	Sales
205	Roshan Lama	36000	Marketing

- Create a database named CollegeDatabase.
- Create a table **Student** and **Employee** with the following fields.
- Insert first row of the table student.
- Insert multiple row of table employee.
- Write SQL statements to add “DOB” with appropriate datatype.
- Write SQL statements to select all the employee whose salary is less than 40000.
- Delete a record from student whose studentid is 101.
- Write SQL statements to show the record data name from table student whose age is 19.
- Write SQL statements to remove the department column from employee table.
- Write SQL statements to delete table employee.

Best Of Luck